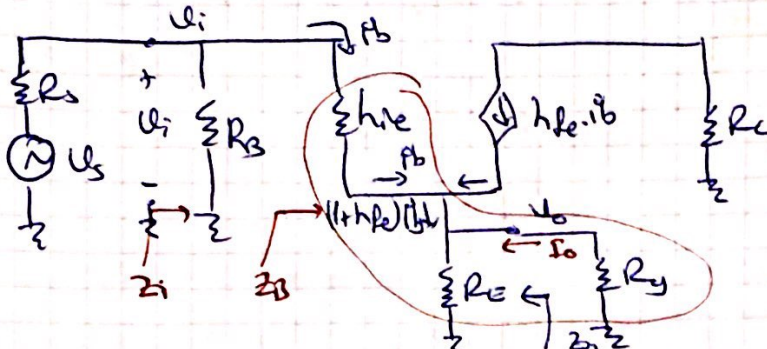


L. EMİTÖR GIKLI YÜKSELTEÇ:

- hoe yok:



① Z_B

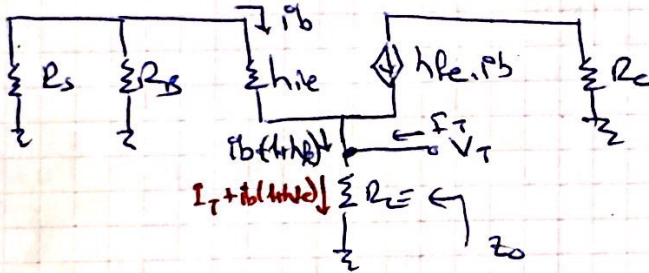
$$Z_B = \frac{u_i}{i_b} \Rightarrow \frac{u_i}{i_b} = h_{ie} + (R_E \parallel R_L) \cdot (1 + h_{fe})$$

$$Z_B = h_{ie} + (R_E \parallel R_L) (1 + h_{fe})$$

$$Z_i = R_B \parallel Z_B$$

② Z_o

U_S kısa devre :



$$V_T = R_E (I_T + i_b (1 + h_{fe})) \quad Z_o = V_T / I_T$$

I_T üzerinden yazılmalı. $\rightarrow$ Güçte gerecek. R_E .

Gerecek:

$$((R_S \parallel R_B) + h_{ie}) \cdot i_b + R_E (I_T + i_b (1 + h_{fe})) = 0$$

$$i_b \left[((R_S \parallel R_B) + h_{ie}) + R_E (1 + h_{fe}) \right] = - R_E I_T$$

A dersen

$$i_b = - \frac{R_E}{A} I_T \Rightarrow V_T = R_E \left(I_T - \frac{R_E}{A} I_T (1 + h_{fe}) \right)$$

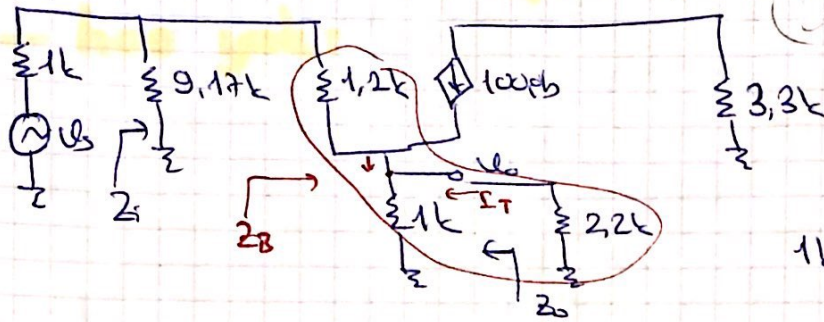
$$V_T = I_T \left(R_E - \frac{R_E^2}{A} (1 + h_{fe}) \right) \Rightarrow Z_o = \left[R_E - \frac{R_E^2}{A} (1 + h_{fe}) \right]$$

③ A_v

$$\left. \begin{aligned} u_o &= (1 + h_{fe}) \cdot i_b \cdot (R_E \parallel R_L) \\ u_i &= i_b \cdot Z_B \end{aligned} \right\} A_v = \frac{(1 + h_{fe}) (R_E \parallel R_L)}{Z_B}$$

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Örnek:



$$1k // 2.2k = 0.68k$$

$$Z_B = 1.2k + (1 + h_{fe}) \cdot (R_E // R_y) = 1.2k + 101 \cdot 0.68k = 70.63k$$

$$Z_i = 9.17k // 70.63k = 8.12k \quad \boxed{Z_o = 8.12k}$$

$$V_T = 1k \cdot (101 I_B + I_T) \Rightarrow \text{girişte cıkıste:}$$

$$\begin{aligned} & \left(\frac{1k // 9.17k}{2.10k} + 1.2k \right) \cdot I_B + 1k \cdot (101 I_B + I_T) = 0 \\ & 2.10k \cdot I_B + 101k \cdot I_B + 1k I_T = 0 \\ & I_B = - \frac{1k}{103.10k} I_T \end{aligned}$$

$$\frac{V_T}{I_T} = 1k \left(- \frac{I_T}{103.10} \cdot 101 + I_T \right) / I_T$$

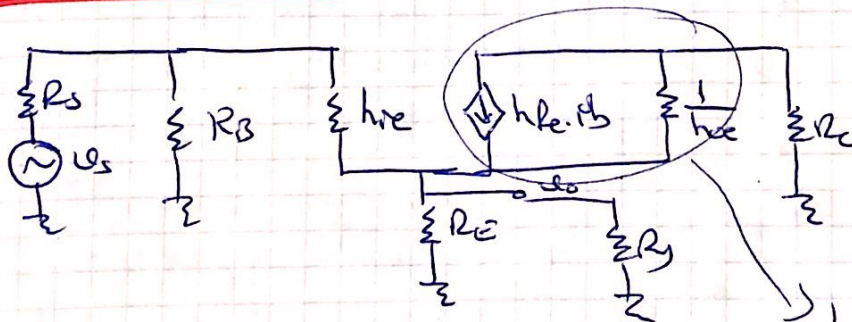
$$\Rightarrow \boxed{Z_o = 20.36 \Omega}$$

$$\begin{aligned} V_o &= 101 I_B \cdot (1k // 2.2k) = 68.68k \cdot I_B \\ V_i &= I_B \cdot Z_B = 70.63k \cdot I_B \end{aligned} \quad \boxed{A_v = 0.97}$$

$$A_{v_s} = 0.97 \cdot \frac{8.12k}{9.12k} = 0.86$$

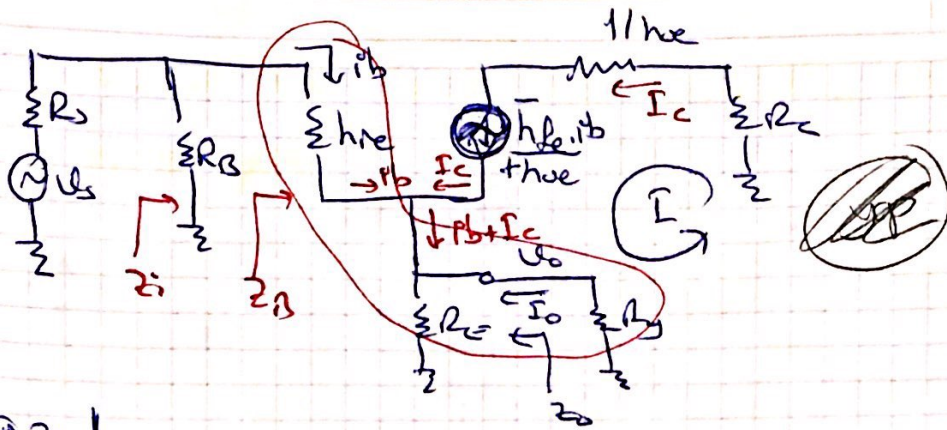
$$A_i = -A_v \cdot \frac{Z_i}{R_y} = -0.97 \cdot \frac{8.12k}{2.2k} = -3.58$$

hoe verken:



kaynak
dönüştürme
yapılır

9



⑧ Zi

$$Z_B = \frac{U_i}{I_B}$$

$$U_i = I_B \cdot h_{ie} + (R_E \parallel R_L) (I_B + I_C)$$

... I_B üzerinden bulunmak gerekiyor. bu bir rim çıkışta cevre yazılması (I noluccevre)

Cevre denk. den

$$\left(R_C + \frac{1}{h_{oe}}\right) I_C - \frac{h_{fe}}{h_{oe}} I_B + (R_E \parallel R_L) (I_B + I_C) = 0$$

Buradan $I_C = A \cdot I_B$ şeklinde bir bağlantı bulunur.

$$U_i = I_B \cdot h_{ie} + (R_E \parallel R_L) (I_B + A \cdot I_B)$$

$$\Rightarrow Z_B = h_{ie} + (R_E \parallel R_L) (1 + A)$$

$$Z_i = Z_B \parallel R_1 \parallel R_2$$

⑨ A_v

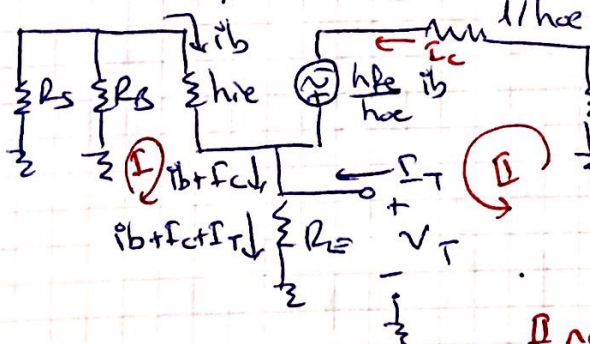
$$U_o = (I_B + I_C) \cdot (R_E \parallel R_L) = I_B (1 + A) (R_E \parallel R_L)$$

$$U_i = I_B \cdot Z_B$$

$$\Rightarrow A_v = \frac{(1 + A) (R_E \parallel R_L)}{Z_B} \parallel$$

⑩ Z_o

V_T / I_T yöntemi (U_s kısa devre)



$$V_T = R_E (I_B + I_C + I_T)$$

I noluccevre:

$$\left[(R_1 \parallel R_2) + h_{ie} \right] \cdot I_B + R_E (I_B + I_C + I_T) = 0$$

$$R_A \Rightarrow I_C + I_T = - (R_A + R_E) \cdot I_B$$

I noluccevre:

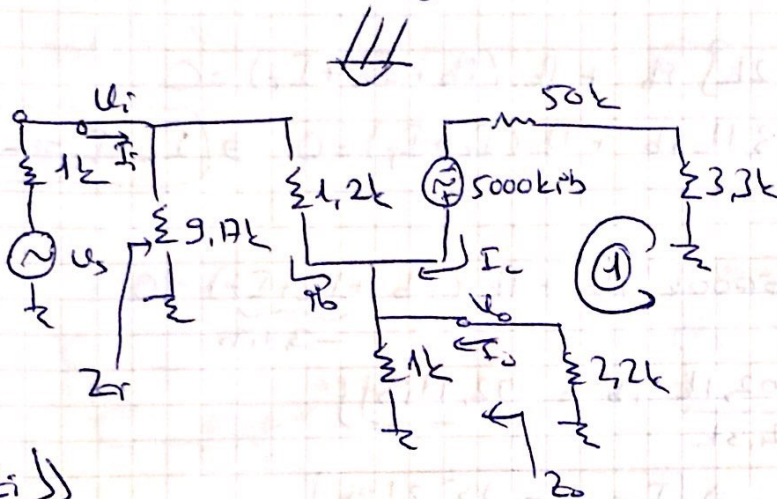
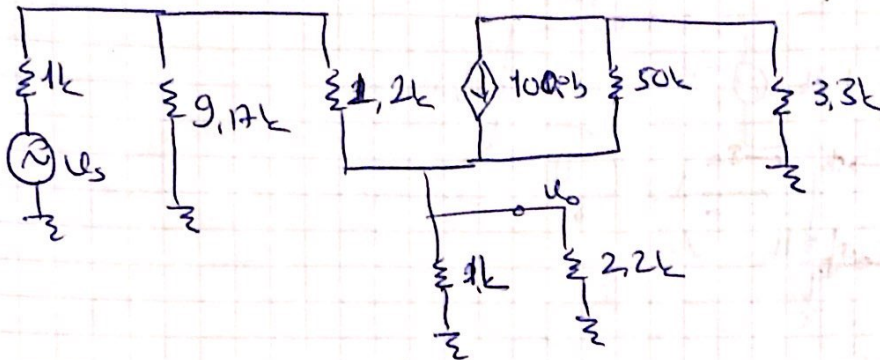
$$\left(R_C + \frac{1}{h_{oe}}\right) I_C - \frac{h_{fe}}{h_{oe}} I_B + R_E (I_B + I_C + I_T) = 0$$

$$- (R_A + R_E) \cdot I_B$$

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ve $I_C = - I_B$ üzerinden bulunur.

Örnek: $h_{fe} = 200$



$$1k \parallel 2,2k = 0,68k$$

⊛ Z_i

I_B ve I_C arasındaki bağlantı için 1 no'lucu denkleme:

$$53,3k \cdot I_C - 5000k I_B + 0,68k \cdot (I_B + I_C) = 0$$

$$I_C = \frac{5000,68k I_B}{54,18k} \approx 92,3 I_B$$

$$Z_B = \frac{U_i}{I_B} \quad \begin{aligned} U_i &= 1,2k I_B + 0,68k (I_B + 92,3 I_B) \\ U_i &= 64,64k I_B \end{aligned}$$

$$Z_B = 64,64k\Omega$$

$$Z_i = 9,17k \parallel 64,64k$$

$$Z_i = 8,03k\Omega$$

⊛ A_u

$$\begin{aligned} U_o &= 0,68k \cdot (I_B + 92,3 I_B) = 63,44k \cdot I_B \\ U_i &= 64,64k I_B \end{aligned} \quad \text{idi}$$

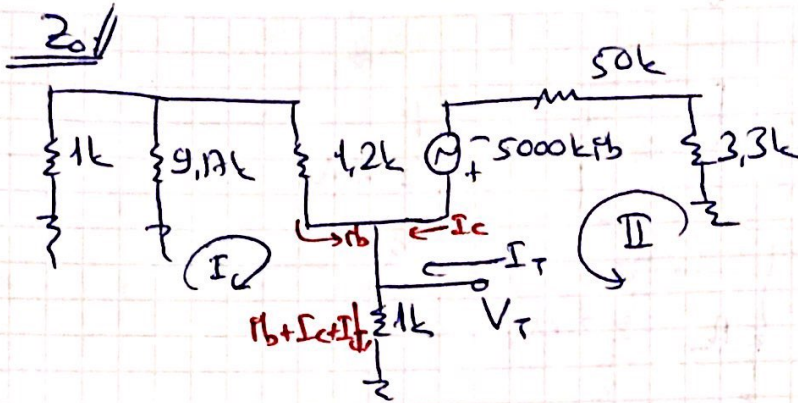
$$\Rightarrow A_u = 0,98$$

$$A_{u_s} = 0,98 \times \frac{8,03k}{9,03k} = 0,87$$

$$A_i = -A_u \cdot \frac{Z_i}{R_s} = -0,98 \cdot \frac{8,03k}{2,2k} = -3,58 = A_{i_s}$$



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I no'lu aqre:

$$[(1k // 9,13k) + 1,2k] I_b + 1k \cdot (I_b + I_c + I_T) = 0$$

$$3,1k I_b + 1k (I_c + I_T) = 0 \Rightarrow \boxed{I_c + I_T = -3,1 I_b}$$

II no'lu aqre:

$$53,3k I_c - 5000k I_b + 1k (I_b + \underbrace{I_c + I_T}_{-3,1 I_b}) = 0$$

$$I_c = \frac{5002,1k}{54,3k} I_b = 92,11 I_b$$

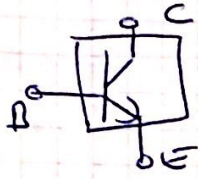
$$I_c + I_T = -3,1 I_b \Rightarrow \boxed{I_T = -95,21 I_b}$$

$$V_T = 1k (I_b + \underbrace{I_c + I_T}_{-3,1 I_b}) \Rightarrow \boxed{V_T = -2,1k I_b}$$

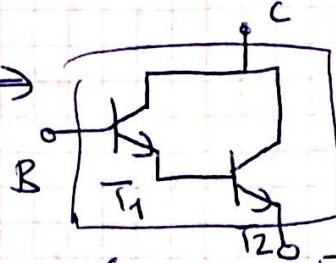
$$\Rightarrow Z_0 = \frac{-2,1k I_b}{-95,21 I_b} = \boxed{22,05 \Omega = Z_0}$$

5. DARLINGTON ÇİFTİ.

Darlington çifti yüksek giriş direnci, yüksek kazançları ve düşük çıkış direnci sağlamak için kullanılır.



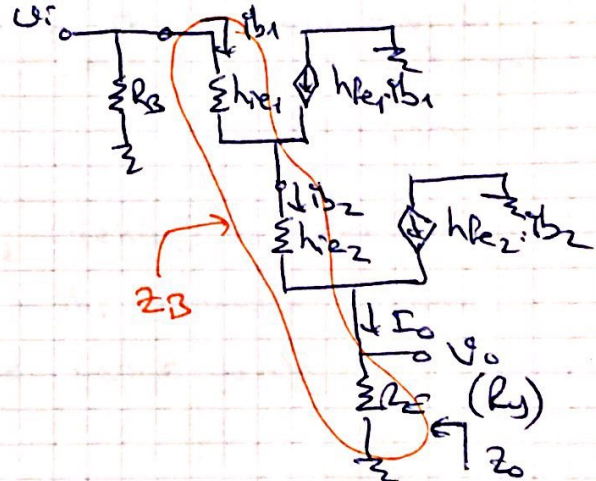
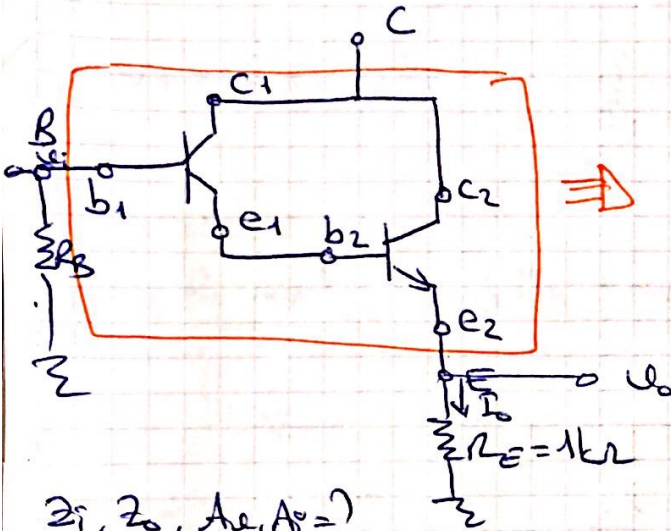
Verne $\Rightarrow$



Kullanılır.

Not: Yük E'ye bağlanır. (emitter çıkışı yüklenir.)

ANALİZ:



$Z_i, Z_o, A_v, A_p = ?$

$$i_{b2} = (1 + h_{fe1}) \cdot i_{b1} \quad ; \quad I_o = (1 + h_{fe2}) i_{b2}$$

$$\Rightarrow I_o = (1 + h_{fe1})(1 + h_{fe2}) i_{b1}$$

$$Z_B = \frac{U_i}{i_{b1}} \Rightarrow U_i = h_{ie1} \cdot i_{b1} + h_{ie2} \cdot i_{b2} + R_E \cdot I_o$$

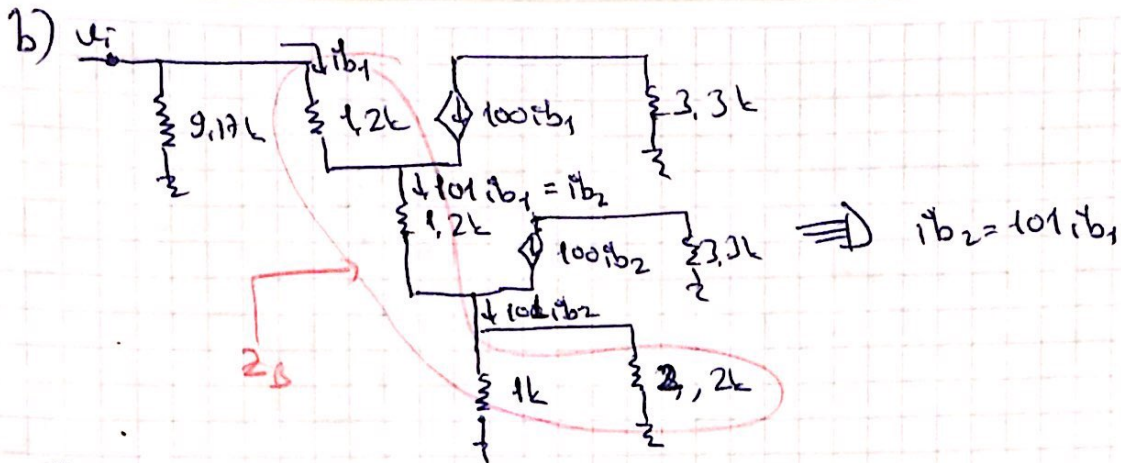
$$U_i = h_{ie1} i_{b1} + h_{ie2} \cdot (1 + h_{fe1}) i_{b1} + R_E (1 + h_{fe1})(1 + h_{fe2}) \cdot i_{b1}$$

$$\Rightarrow Z_B = \frac{U_i}{i_{b1}} \Rightarrow Z_B = h_{ie1} + h_{ie2}(1 + h_{fe1}) + R_E(1 + h_{fe1})(1 + h_{fe2})$$

$$(Z_i = R_B \parallel Z_B) \parallel$$

$$Z_o = \frac{U_o}{I_o} = \frac{V_o}{I_o} = R_E$$

$$A_v = \frac{I_o}{I_p} = \frac{(1 + h_{fe1})(1 + h_{fe2}) i_{b1}}{i_{b1} + \frac{U_i}{R_B}} \quad \text{den bulunur.}$$



$$Z_B = \frac{U_i}{i_{b1}}$$

$$U_i = 1,2k i_{b1} + 1,2k (101 i_{b1}) + (1k \parallel 2,2k) \cdot 101 i_{b2}$$

$$U_i = 7058,4k i_{b1} \Rightarrow Z_B = 7,06 M\Omega$$

$$Z_i = 9,17k \parallel 7,06M = 9,15k\Omega$$

$$A_v = \frac{U_o}{U_i}$$

$$U_o = 101 i_{b2} \cdot 0,68k = 6936,68k i_{b1}$$

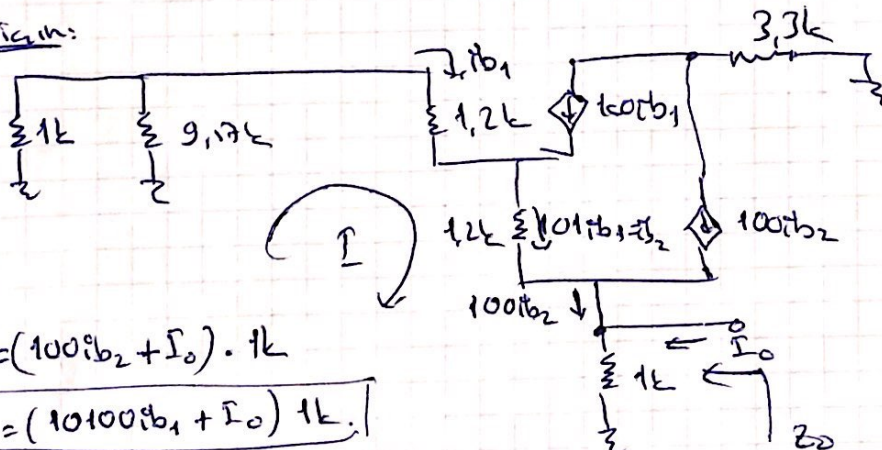
$$U_i = 7058,4k i_{b1}$$

$$\Rightarrow A_v = 0,98$$

$$A_i = -A_v \cdot \frac{Z_i}{R_y}$$

$$A_i = -0,98 \cdot \frac{9,15k}{2,2k} = -4,07$$

Z_o için:



$$V_T = (100 i_{b2} + I_o) \cdot 1k$$

$$V_T = (10100 i_{b1} + I_o) 1k$$

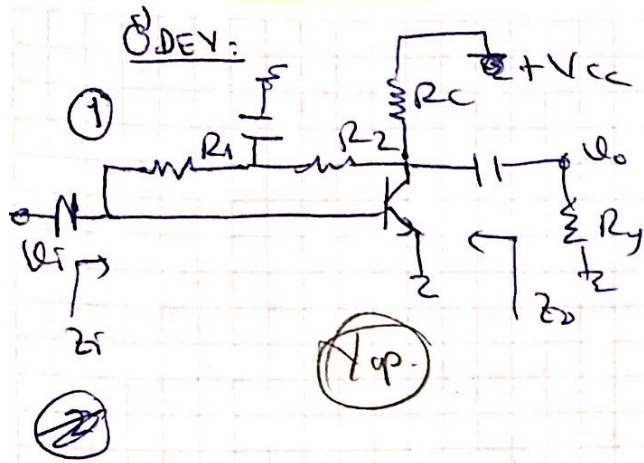
$$[(1k \parallel 9,17k) + 1,2k] \cdot i_{b1} + 1,2k \cdot 101 i_{b1} + 1k \cdot (10100 i_{b1} + I_o) = 0$$

$$i_{b1} = -\frac{1}{10223,3} I_o$$

$$\Rightarrow V_T = 12,10 I_o$$

$$Z_o = 12,1\Omega$$

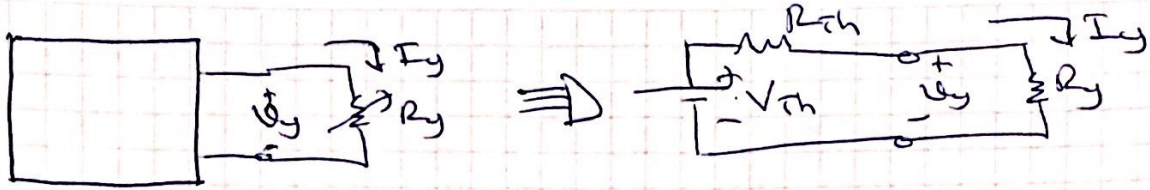
(1k)



a) AC ezdegeri ni c.2

b) $R_1 = 100k$ $R_2 = 100k$
 $R_C = 2,2k$ $R_E = 2,2k$
 $R_L = 2,2k$ $R_3 = 2,2k$
 $h_{ie} = 1,2k$ $h_{fe} = 100$
 r_{in} Z_i , Z_o , A_v , $A_v = ?$

6. Maksimum Güç Aktarımı Teoremi:

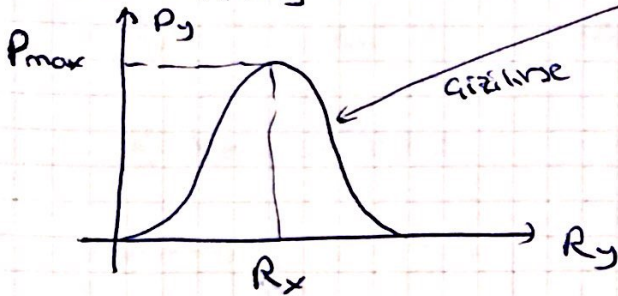


$R_L = 0$ olursa kısa devre
 $R_L = \infty$ olursa açık devre.

$$P_L = V_L \cdot I_L = I_L^2 \cdot R_L$$

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

$$\Rightarrow P_L = \left(\frac{V_{th}}{R_{th} + R_L} \right)^2 R_L$$



maksimum güç için $R_L = R_x$ olmalı.

$$\frac{dP_L}{dR_L} = 0 \text{ olan nokta.}$$

$$\frac{d}{dR_L} \left(\frac{V_{th}^2 R_L}{(R_{th} + R_L)^2} \right) = \frac{V_{th}^2 (R_{th} + R_L)^2 - V_{th}^2 R_L \cdot 2(R_{th} + R_L)}{(R_{th} + R_L)^4} = 0$$

$$(R_{th} + R_L)^2 = 2R_L(R_{th} + R_L) \quad \left\{ \begin{array}{l} R_L = R_{th} \text{ çıkar.} \end{array} \right.$$

Yani maksimum güç için $R_L = R_{th}$ olmalı. $\downarrow$

Genel olarak :

$$Z_L = Z_{th}^*$$

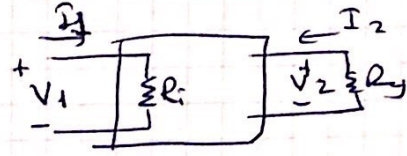
$$Z_L = R_{th} - jX_{th} \quad (Z_{th} = R_{th} + jX_{th})$$

7. Desibel Kavramı

Logaritmik kazanç.

P_2 = çıkış gücü , P_1 = giriş gücü olmak üzere;

$$G_{dB} = 10 \log \frac{P_2}{P_1}$$

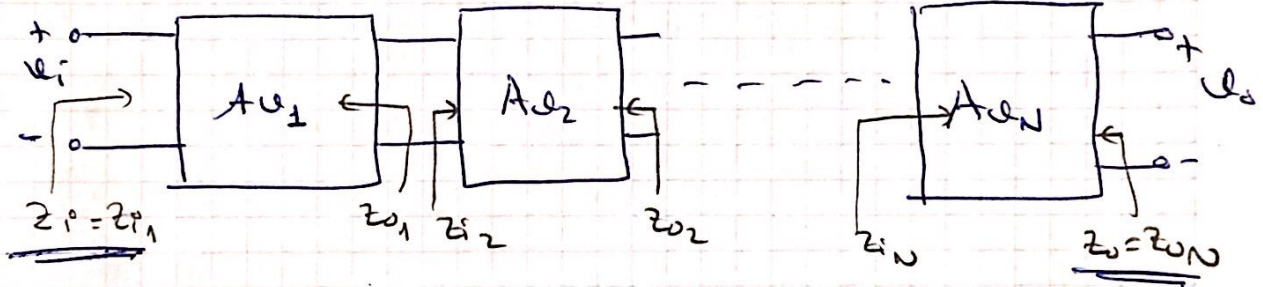


$$G_{dB} = 10 \log \frac{V_2^2 / R_j}{V_1^2 / R_i}$$

$R_j = R_i$ ise :

$$G_{dB} = 20 \log \frac{V_2}{V_1} \quad (dB)$$

Kaskad Sistemler:



$$A_v = \frac{V_o}{V_i} = A_{v_T}$$

$$A_{v_T} = A_{v_1} \cdot A_{v_2} \cdot A_{v_3} \cdot \dots \cdot A_{v_n} \cdot K$$